

Exercise Sheet

Task 1 – Fibonacci Sequence

The Fibonacci sequence is a numerical series where each number in the sequence is decided by adding the two previous numbers from it.

Let $F(n)$ be the n th Fibonacci number, then $F(n) = F(n - 2) + F(n - 1)$
Additionally $F(0) = 0$ and $F(1) = 1$

Implement a program that computes the n th Fibonacci number, where n is an integer in register t0. The return value should also be put into register t0. If the number in register t0 is negative, your program should return -1. Make sure to include an overflow check and comments.

- a) Draw a Flow Chart to plan your implementation
- b) Implement the Algorithm